METRIC

A-A-52424A
October 21, 1994
SUPERSEDING
A-A-52424
February 2, 1993
(See 5.4)

COMMERCIAL ITEM DESCRIPTION

CAPS, RADIATOR: PRESSURE RELIEVING (METRIC)

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. Scope. This CID covers radiator pressure caps with a nominal pressure rating of 48 kPa. Caps covered by this CID are designed for use with industrial and automotive internal-combustion engine radiators.

1.1 Classification. Radiator caps covered by this CID are listed as follows:

Type	Design	Cap size
1	without handle	(64 mm)
2	with handle	(95 mm)

2. SALIENT CHARACTERISTICS

2.1 Materials. The materials used in the radiator cap shall be made of corrosion resistant steel with exception to the handle, crown, hook and chain which can be made of zinc plated steel per ASTM B633. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.5).

2.2 Chain and hook. Radiator cap shall be supplied with chain (see ASTM A466 class SF) and hooks. Hook design is optional but shall be easy to hook and remove without damage to the radiator cap, chain or hook (see figures 1 and 2).

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-T, Warren, MI 48397-5000.

FSC 2930

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2.3 Compatibility. Radiator caps shall connect to standard filler necks conforming to the following:

Type	Standard filler neck
1	MS35841
2	MS51063

2.4 Design and performance. Caps shall meet the requirements of this CID when used with antifreeze/coolant. Design shall be as specified in figures 1 and 2.

2.4.1 Fluid resistance. Gasket shall show no evidence of deterioration, flaking, change in volume (not more than 3%), or coolant discoloration when immersed in coolant such as antifreeze or alcohol for 158 hours.

2.4.2 Corrosion resistance. Caps shall conform to 2.4.3, 2.4.4 and 2.4.5 after being subjected to a salt spray test for 96 hours in accordance with ASTM B117.

2.4.3 Pressure valve. Pressure valve shall relieve at no less than 45 kPa and no more than 55 kPa at a temperature of 68 degrees Celsius (°C).

2.4.4 By-passing. Pressure valves shall not show evidence of by-passing at pressures up to 43 kPa.

2.4.5 Vacuum valve. The vacuum valve shall relieve at less than 7 kPa vacuum at a temperature of 68°C.

2.5 Extreme low temperature. The cap shall be able to be removed by hand from the stop contact points of the filler neck at the environmental temperature of -54°C using arctic mittens. The torque used to remove the cap shall not be more than 5.6 Nm for type 1 and 14.1 Nm for type 2.

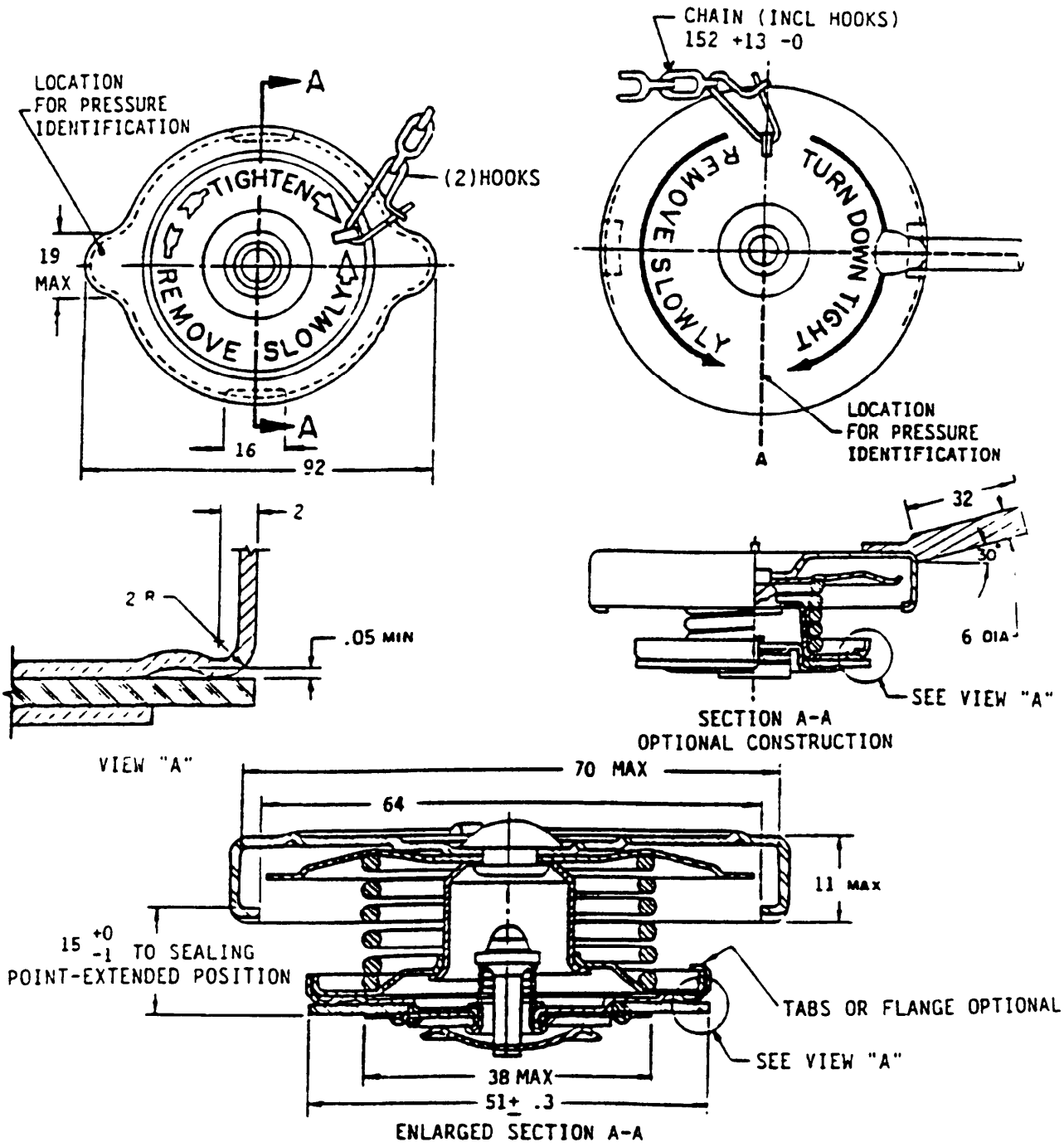
2.6 Identification and marking. Identification and marking of the radiator cap, as a minimum, shall include the part identification number, pressure identification, manufacturer's CAGE code (or equivalent identification), and part number. The radiator cap shall have markings to indicate direction of turn to remove or tighten the cap.

3. QUALITY ASSURANCE PROVISIONS

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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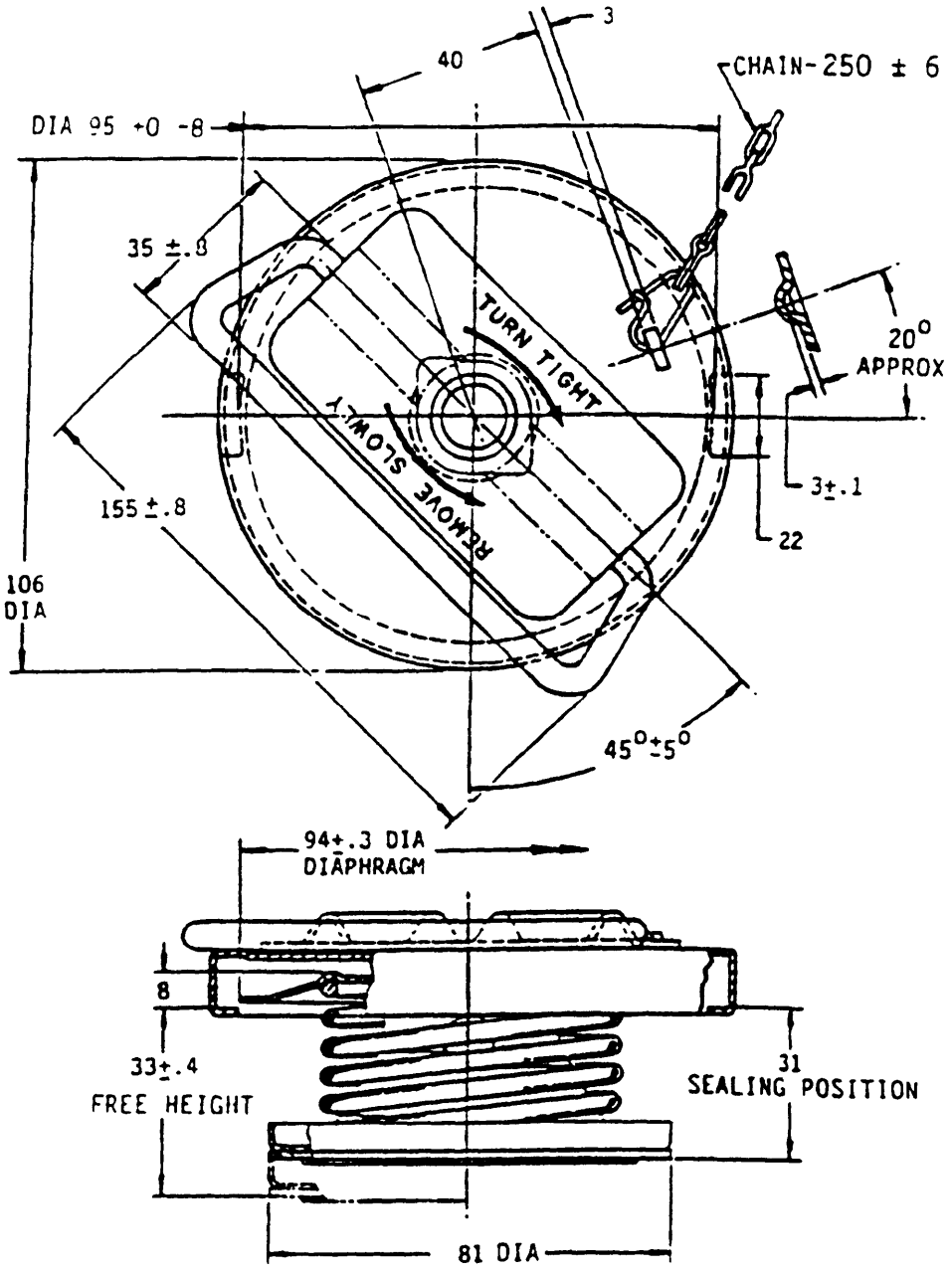


PIN No.	Former MS Part No.
A 52424-1	MS35840-1

NOTE: Dimensions are in millimeters. Tolerance ± .25

FIGURE 1. Radiator cap - type 1.

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PIN No.	Former MS Part No.
A 52424-2	MS51062-1

NOTE: Dimensions are in millimeters. Tolerance ± .25

FIGURE 2. Radiator cap - type 2.

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4. PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order (see 5.2).

5. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

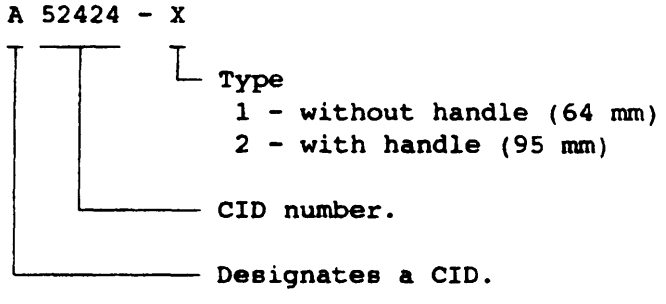
5.1.1 Government specifications and standards. Copies of MS35841, Neck Filler Radiator, and MS51063, Neck Filler Radiator, are available from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.

5.1.2 Non-Government publications. Copies of ASTM A466-91 "Standard Specification Weldless Carbon Steel Chain"; ASTM B117 "Standard Test Method of Salt Spray (Fog) Testing"; and ASTM B633-85 "Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel"; are available from the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of the DODISS to be cited in the solicitation and the specific issue of individual documents referenced.
c. Selection of applicable level and packaging requirements.
d. Part or identification number.

5.3 Part or identifying number (PIN). The PIN to be used for radiator caps acquired to this CID is as follows:



5.4 Cross-reference and supersession data. Radiator caps conforming to this CID are interchangeable/substitutable with radiator caps conforming to MIL-C-45167C, dated 20 April 1988, MS35840D, dated 21 January 1976, and MS51062E, dated 26 March 1979, which are superseded by this CID.

5.5 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

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